

$$a) y(n) = x(n-3) + x(n-2) + x(n-1) + x(n)$$

$$b) y(n) = x(n-4) + x(n-3) + x(n-2) + x(n-1) + x(n)$$

$$c) y(n) = x(n) - x(n-1)$$

$$d) y(n) = x(n) - x(n-2)$$

1)

$$a) T_1(z) = z^0 + z^{-1} + z^{-2} + z^{-3} = 1 + z^{-1} + z^{-2} + z^{-3}$$

$$T_1(z) = \frac{z^3 + z^2 + z^1 + 1}{z^3}$$

$$b) T_2(z) = z^0 + z^{-1} + z^{-2} + z^{-3} + z^{-4} = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4}$$

$$T_2(z) = \frac{z^4 + z^3 + z^2 + z^1 + 1}{z^4}$$

c)

$$T_3(z) = (-1)z^{-1} + z^0 = 1 - z^{-1} = \frac{z-1}{z}$$

d)

$$T_4(z) = (-1)z^{-2} + z^0 = 1 - z^{-2} = \frac{z^2-1}{z^2}$$

2)

$$z = e^{j\Omega} \quad M = \frac{M-1}{2}$$

$$e^{j\theta} + e^{-j\theta} = 2\cos\theta \quad e^{j\theta} - e^{-j\theta} = 2j\sin\theta \quad j = 1e^{j\pi/2}$$

$$M = 3/2 \quad T_1(\Omega) = 1 + e^{-j\Omega} + e^{-2j\Omega} + e^{-3j\Omega} = e^{-j\frac{3}{2}\Omega} (e^{j\frac{3}{2}\Omega} + e^{j\frac{1}{2}\Omega} + e^{-j\frac{1}{2}\Omega} + e^{-j\frac{3}{2}\Omega})$$

$$= [2\cos(\frac{\Omega}{2}) + 2\cos(\frac{3\Omega}{2})] \cdot e^{-j\frac{3}{2}\Omega}$$

$$T_{R1} = 2\cos(\frac{\Omega}{2}) + 2\cos(\frac{3\Omega}{2}) \Rightarrow |T| = |T_{R1}| ; \angle T = -\frac{3}{2}\Omega$$

b)

$$M = 2 \quad T_2(\Omega) = 1 + e^{-j\Omega} + e^{-2j\Omega} + e^{-3j\Omega} + e^{-4j\Omega} = e^{-j2\Omega} (e^{j2\Omega} + e^{j\Omega} + e^0 + e^{-j\Omega} + e^{-2j\Omega})$$

$$T_2(\Omega) = [1 + 2\cos(\Omega) + 2\cos(2\Omega)] e^{-j2\Omega}$$

$$T_{R2} = 1 + 2\cos(\Omega) + 2\cos(2\Omega) \Rightarrow |T| = |T_{R2}| ; \angle T = -2\Omega$$

c) $M = \frac{1}{2}$

$$T_3(\Omega) = 1 - e^{-j\Omega} = e^{-j\frac{1}{2}\Omega} (e^{j\frac{1}{2}\Omega} - e^{-j\frac{1}{2}\Omega}) = [2j\sin(\frac{1}{2}\Omega)] e^{-j\frac{1}{2}\Omega}$$

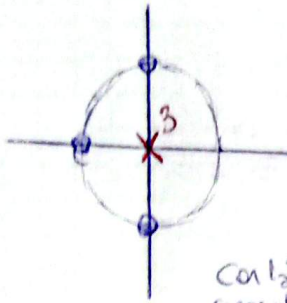
$$\Rightarrow T_3(\Omega) = 2\sin(\frac{1}{2}\Omega) e^{-j(\frac{1}{2}\Omega - \pi/2)}$$

$$T_{R3} = 2\sin(\frac{1}{2}\Omega) \Rightarrow |T| = |T_{R3}| ; \angle T = -\frac{1}{2}\Omega + \frac{\pi}{2}$$

d) $M=1$ $T_4(\Omega) = 1 - e^{-j2\Omega} = e^{-j\Omega} (e^{j\Omega} - e^{-j\Omega}) = [2j \sin(\Omega)] e^{-j\Omega}$
 $\Rightarrow T_4(\Omega) = [2 \sin(\Omega)] e^{-j(\Omega - \pi/2)}$
 $T_{H4} = 2 \sin(\Omega) \Rightarrow |T| = |T_{H4}| ; \angle T = -\Omega + \pi/2$

Graficos

a)



$$z^3 + z^2 + z + 1 = 0$$

$$(z+1)(z^2+1) = 0$$

$$z = -1, j, -j$$

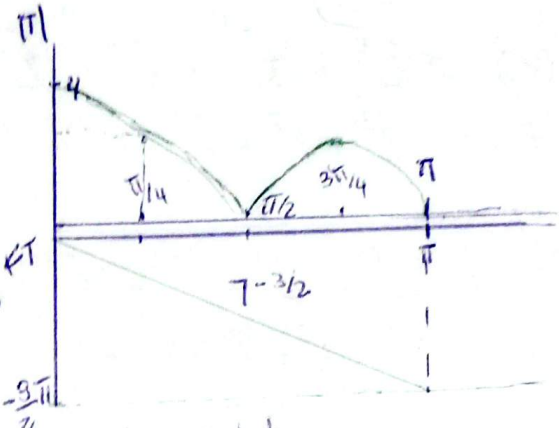
$$T(z) = \frac{1 - z^{-4}}{1 - z} = 0$$

Cont. Geométrica

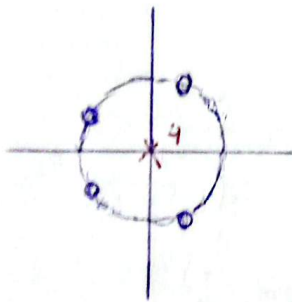
$$e^{-j4\omega} = 1$$

$$4\omega = 2\pi k$$

$$\omega = \frac{\pi}{2} k \rightarrow k = 1, 2, 3$$



b)



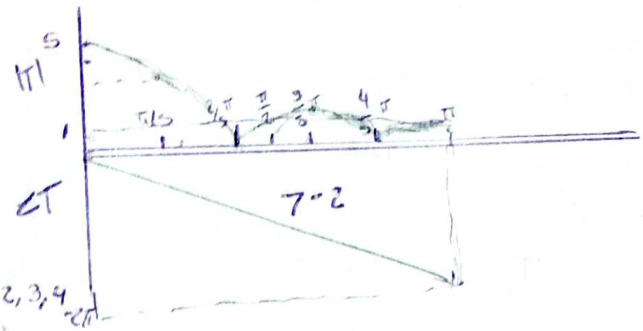
$$T(z) = \frac{1 - z^5}{1 - z}$$

$$e^{-j5\omega} = 1$$

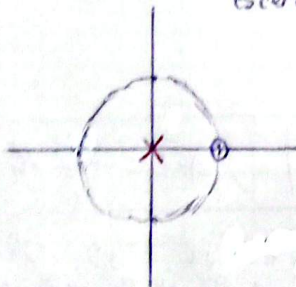
$$5\omega = 2\pi k$$

$$\omega = \frac{2\pi}{5} k \quad k = 1, 2, 3, 4$$

$k=0 \rightarrow$ No Cero



c)

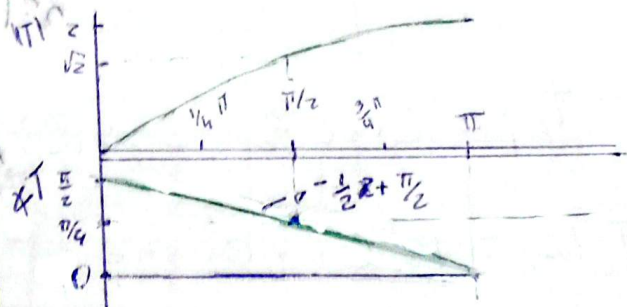


$$1 - e^{-j\omega} = 0$$

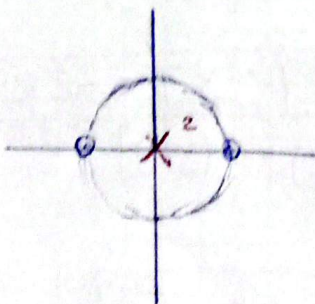
$$e^{-j\omega} = 1$$

$$\omega = 2\pi k$$

$$\omega = 2\pi k \rightarrow k = 0$$



d)



$$z^2 - 1 = 0$$

$$z = \pm 1$$

$$1 - e^{-j2\omega} = 0$$

$$e^{-j2\omega} = 1$$

$$2\omega = 2\pi k$$

$$\omega = \pi k \quad k = 0, 1$$

